Cupcake Tycoon: the full game document

Build Queue 400 (2026-10-27). Her words: "give me the full doc explaining all the rules, design, game play, etc for cupcake tycoon".

This describes the game exactly as it is built today, then the open plan.

1. What it is

- An idle bakery empire game. The student starts with one cupcake counter and grows a bakery with eight counters, more ovens, bakers who work alone, shop stars and a cake display case.
- It is on purpose different from Diner Dash. Diner Dash is time management: cook and serve each customer before they leave. Cupcake Tycoon has no customers to serve. The student bakes, grows and upgrades.
- Questions come from her starred question sets (or the student's open gameplay assignment first), on the same question screen as every game.

2. Where it lives

- A card in the game picker (Arcade), and a "Cupcake Tycoon" building role in Build Mode (give it to a bakery or farm stand in Town Square).
- Route `/student/cupcake-tycoon`; code `src/routes/student/IdleTycoon.tsx`; art `public/games/cupcake-tycoon/` (her pixel dessert packs, Build Queue 289).
- Saved per student in the `tycoon:<studentId>` row: shop coins, shop coins earned in all, and each counter's ovens and baker.

3. The two kinds of money

- **Shop coins (🪙):** pretend money inside the game. They only buy ovens, counters and bakers. They never turn into Class Cash or prizes.
- **Class Cash:** $1 for every right answer, paid once when the student leaves (the same as every game), with the coin drop and a teacher report.

4. How it plays, rule by rule

The big cupcake. Tap it to bake one by hand: 1 shop coin a tap (3 during a Golden Whisk).

Counters. Each counter bakes one treat. Tap a counter to start a batch; a bar fills, and the batch sells for its price times the number of ovens.

Counter	Opens at (shop coins)	Batch time	Sells for (each oven)	First extra oven	Baker

Cupcakes	open at the start	2 s	1	5	40
Cookies	60	4 s	8	60	400
Donuts	600	6 s	60	600	3,000
Pies	5,000	9 s	400	5,000	25,000
Cakes	40,000	14 s	3,000	40,000	200,000
Waffles	300,000	18 s	22,000	300,000	1,500,000
Pancakes	2,500,000	22 s	160,000	2,500,000	12,000,000
Rainbow Cakes	20,000,000	28 s	1,200,000	20,000,000	90,000,000

More ovens. Each oven makes that counter's batch bigger by one batch price. Each new oven costs 15% more than the one before (rounded).

Bakers. Hiring a baker makes that counter start its next batch by itself, forever.

Golden Whisk. Tap "Answer for a Golden Whisk": one question. A right answer makes every counter, and the big cupcake, three times as fast for 30 seconds. A wrong answer shows the right answer and gives no whisk. The button waits while a whisk is running.

Shop stars. One to five stars at 0, 100, 1,000, 10,000 and 100,000 shop coins earned in all. The sign shows "X / Y shop coins for the next star", then "Five star bakery empire!".

Cake display case. One of her big pixel cakes appears for each milestone of shop coins earned in all: 50, 300, 1,500, 8,000, 40,000, 200,000, 1 million, 5 million, 25 million and 100 million (Birthday, Strawberry Pudding, Blueberry, Chocolate, Orange, Bubblegum, Brownie, Maple Popcorn, Double Chocolate, Jelly).

While away. Counters with a baker keep earning while the student is gone, up to 10 minutes' worth. On return, a "Welcome back!" card says how many shop coins the bakers earned.

Number display. Under 10,000 shows the whole number; 10,000 and up shows K (thousands); a million and up shows M with one decimal.

5. Shared parts (the same as every game)

- The settings card ("questions each time"), the ⚙️ button, and assignment locks.
- One try per question; a wrong answer moves to a new question; Listen reads it with a human voice; tools (calculator, scratchpad, text size, highlighter).
- 🏆 My best (shop coins earned in a visit), sounds with an off switch, "Leave this game?", the ☰ main menu.
- Every right answer counts toward the Daily Streak and is logged for the teacher.

6. Design

- Bakery look with her pixel art (Dessert Icons and Bakery packs); each counter shows its treat; the PixCon title font; a gold glow on the screen during a Golden Whisk.
- iPad first: big tap targets, no hover, works in both orientations.

7. Claudia's review of what is holding it back (from Build Queue 208)

- Questions are optional and rare: only the Golden Whisk asks one, so a student can play a whole period without answering.
- Tap mashing: the early game rewards fast tapping (about 60 taps to open Cookies).
- It looks like a list, not a bakery: no ovens, bakers or customers to watch.
- No goal on screen telling the student what to aim for next.
- Big numbers (K, M) shown without help.
- Little read-aloud: counters, prices and the Welcome back card do not speak.
- No long game after the last counter and five stars.
- Shop coins go nowhere (built as no, on purpose).

8. The plan (not built yet, waiting on her)

- **Phase 1, questions in the play loop:** a question to open a counter or hire a baker; a Rush order about every 90 seconds ("Rosa needs 12 cookies for a party!") for a big bonus; the Golden Whisk with a short cooldown; focus sets first.
- **Phase 2, a clear next step:** a goal card with 🔊; hold to mix instead of tap mashing; number words under big prices; 🔊 on every counter.
- **Phase 3, a bakery you can see:** glowing ovens, trays sliding into the case, bakers, customers who wave (decoration only), her Style character at the register; oven ding, mixer and register sounds (calm in Reduced motion).
- **Phase 4, the long game:** new bakeries at five stars (shop coins start over, every counter earns double), a recipe book of flavors, a daily special, a teacher view.

9. Her answers needed (Build Queue 208)

- Should opening a counter and hiring a baker ask a question? (Suggested: yes.)
- Rush orders: how often? (Suggested: about every 90 seconds.)
- Should shop coins ever turn into Class Cash or prizes? (Suggested: no, but five stars could pay a one-time reward.)
- Hold to mix instead of tapping the big cupcake? (Suggested: yes.)
- Which phase first? (Suggested: Phase 1, then 2, then 3.)